

UNIFIED MASTER TELEMETRY GLOSSARY & METRIC SPECIFICATION

MyceliaLM Cognitive Architecture & MASSIF Latent Trajectory Framework
Continuous SDE Kinematics, Closed-Loop Governors, Pressure Tensor Diagnostics, and Macroscopic Phase Transitions

Daniel Solis – Principal Investigator, Dubito Inc. / Ergo Sum AGI Safety Systems

September 2026 – Version 3.5 Unified Synthesis

Abstract

This document serves as the unified, mathematically complete, and rigorous master glossary synthesizing all parameters, observables, control variables, and diagnostic metrics from the **MyceliaLM Telemetry Dashboard Parameter Glossary** and the **MASSIF (Multiscale Attractor Stability & Stress Inference Framework) Telemetry Glossary**. It bridges continuous-time Itô Stochastic Differential Equation (SDE) trajectory kinematics, Fokker-Planck jump diagnostics, depth-dependent friction cascades, closed-loop cybernetic governor stacks, pressure tensor thermodynamics, and Kuramoto-inspired order parameters. Every metric is defined with its exact mathematical expression, plain-language translation, target/healthy operational ranges, and diagnostic significance for ex-ante AI safety.

Contents

1	Modification and Addition Log	2
2	Foundational Hidden-State Geometry & Governor Architecture	3
2.1	Hidden-State Trajectory Geometry	3
2.2	The Four-Loop Closed-Loop Governor Stack	3
3	Stochastic Differential Equation (SDE) & Fokker-Planck Diagnostics	4
4	Latent Trajectory Kinematics & Differential Geometry	5
5	Friction, Depth Variance, & Silicon Hardware Rails	7
6	Attention Consensus, Inter-Head Fiber Curvature, & Spectral Dynamics	8
7	Instability Forecasting & Closed-Loop Control	9
8	Pressure Tensor (II) & Macroscopic Order Parameters	10
9	Kuramoto Synchronization & Geometric Convergence (GCI)	11
10	Advanced Anisotropic & Subspace Metrics	12
11	Summary Matrix of Telemetry Observables	12

1 Modification and Addition Log

Comprehensive Audit: Merging MyceliaLM and MASSIF Sources

This master document unifies two distinct telemetry files: the operational dashboard specification (*MyceliaLM Telemetry Dashboard*) and the theoretical trajectory analysis specification (*MASSIF Telemetry Glossary*). The following synthesis rules, additions, and modifications were applied:

1. **Unification of Terminology & Notation:** Standardized notation across both files. Hidden residual states are defined as $h_t^{(l)} \in \mathbb{R}^d$, velocity as $v_t^{(l)}$, persistence as I_t , and optimization pressure as Π .
2. **Integration of SDE/Fokker-Planck Diagnostics:** Merged Pawula higher-moment observables ($D^{(1)}, D^{(2)}, D^{(4)}, \mathcal{R}_{\text{Pawula}}$) from MASSIF with the real-time SDE Signal-to-Noise Ratio (ρ_{dir}) from MyceliaLM.
3. **Added Missing Metrics from MASSIF & GitHub Repositories:**
 - **Dubito Index / Paradox Score:** Added explicit paradox/uncertainty score metric ($\text{Dubito} > 7.0$) monitoring multi-head dissensus and trajectory curvature before confidence collapse.
 - **Effective Warning Lead Time (τ_{eff}):** Added the positive predictive warning window (+32.4 tokens) quantifying lead time over output probability metrics.
 - **Cache-Fused Kinematic Rail (CFKR) Friction Delta (Δ_{CFKR}):** Added in-vivo KV-cache mutation metric grafting early-to-late variance differentials.
 - **Block-4 Fourier Spectral Resonance ($S(\omega)$):** Added spectral prompt susceptibility metric measuring frequency-selective entrainment.
 - **Kuramoto Synchronization Velocity (V_t) & Acceleration (A_t):** Added derivatives of the macroscopic order parameter for dynamical taxonomy classification.
4. **Four-Fold Standardized Structure per Metric:** Every entry now explicitly includes:
 - (a) **Definition:** What the metric represents geometrically and physically.
 - (b) **Mathematical Calculation & Plain Words:** Formal equation and word-by-word explanation.
 - (c) **Target / Healthy Values:** Explicit quantitative boundaries, warning thresholds, and failure regimes.
 - (d) **Diagnostic Significance:** Why the metric matters for real-time safety, governance, and model performance.

2 Foundational Hidden-State Geometry & Governor Architecture

2.1 Hidden-State Trajectory Geometry

Metric 1.1: Hidden State Residual Vector ($h_t^{(l)}$)

What it is: The continuous representational state vector in the residual stream evolving through network layers $l \in [1, L]$ and autoregressive time steps $t \in [1, T]$.

Calculation:

$$h_t^{(l)} \in \mathbb{R}^d, \quad l \in \{1, \dots, L\}, \quad t \in \{1, \dots, T\}$$

In Words: "The hidden state at layer l and timestep t is a vector belonging to the d -dimensional real vector space $\mathbb{R}^d$, evolving sequentially across layers l from 1 to L and generation steps t from 1 to T ."

Target / Healthy Values: Mean Euclidean norm $\|h_t^{(l)}\|_2$ bounded between 18.0 and 24.0 Activation Magnitude Units.

Why it matters: Treats model execution as a continuous Riemannian trajectory rather than a black-box function, enabling real-time differential geometric telemetry during the forward pass.

2.2 The Four-Loop Closed-Loop Governor Stack

Metric 1.2: FFN Veto Governor Norm ($\|h_{\text{FFN}}^{(l)}\|_2$)

What it is: Real-time activation magnitude of the feed-forward network (FFN) sub-layer contribution.

Calculation:

$$\|h_{\text{FFN}}^{(l)}\|_2 \leq \tau_{\text{FFN}}$$

In Words: "The Euclidean two-norm of the feed-forward network hidden state vector h_{FFN} must remain less than or equal to the target threshold τ_{FFN} ."

Target / Healthy Values: Bounded below $\tau_{\text{FFN}} = 12.5$. Governor work triggers when $\|h_{\text{FFN}}\|_2 > 12.5$.

Why it matters: Prevents FFN sub-layers from overwhelming attention representations with high-norm memorized facts or hallucinated noise.

Metric 1.3: Alpha Scaling Governor Norm ($\|\alpha_a a^{(l)} + \alpha_f f^{(l)}\|_2$)

What it is: Combined weighted contribution vector from self-attention ($a^{(l)}$) and FFN ($f^{(l)}$) branches under layer-scaling parameters α_a, α_f .

Calculation:

$$\|\alpha_a \cdot a^{(l)} + \alpha_f \cdot f^{(l)}\|_2 \leq \tau_\alpha$$

In Words: "The Euclidean two-norm of the combined vector sum of attention contribution $\alpha_a a$ plus feed-forward contribution $\alpha_f f$ must be less than or equal to threshold τ_α ."

Target / Healthy Values: Bounded below $\tau_\alpha = 15.0$. Normal operating range: 8.5 – 12.0.

Why it matters: Prevents explosive gain accumulation across sequential transformer layers, stabilizing Manual Muon and AdamW optimization.

Metric 1.4: Soft Cap Governor Norm ($\|r^{(l)}\|_2$)

What it is: Total residual stream vector norm prior to final layer-norm projection.

Calculation:

$$\|r^{(l)}\|_2 \leq \tau_{\text{cap}}$$

In Words: "The Euclidean norm of the residual stream vector r must be less than or equal to the soft-capping threshold τ_{cap} ."

Target / Healthy Values: Bounded below $\tau_{\text{cap}} = 28.0$. Smooth softplus compression activates above 22.0.

Why it matters: Eliminates catastrophic norm explosion and logit overflow without introducing discontinuous gradient clipping.

Metric 1.5: Model Predictive Control (MPC) Governor (I_{combined})

What it is: Predictive instability score calculated by a temporal Kalman/Bayesian forecaster observing trend acceleration.

Calculation:

$$I_{\text{combined}}^{(l)} \leq I_{\text{target}}$$

In Words: "The trend-extended combined instability score I_{combined} at layer l must remain less than or equal to target instability I_{target} ."

Target / Healthy Values: Target threshold $I_{\text{target}} = 0.45$. Warning range: $0.45 - 0.70$. Critical: > 0.70 .

Why it matters: Provides proactive, predictive intervention by executing fractional dampening before trajectory degeneration serializes into output tokens.

3 Stochastic Differential Equation (SDE) & Fokker-Planck Diagnostics

Metric 2.1: First-Order Drift Velocity ($D^{(1)}$ / μ_{drift})

What it is: Deterministic first-order drift vector driving logical progression through latent semantic space under an Itô SDE ($dz_t = D^{(1)}dt + \sqrt{2D^{(2)}}dW_t$).

Calculation:

$$D^{(1)}(z) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \mathbb{E}[\Delta z \mid z_t = z]$$

In Words: "The first-order drift vector $D^{(1)}$ equals the limit as delta t approaches zero of one divided by delta t , multiplied by the expected displacement delta z given state z ."

Target / Healthy Values: Magnitude range 18.79 to 21.58.

Why it matters: Measures the strength of deterministic reasoning drive versus stochastic diffusion noise.

Metric 2.2: Second-Order Diffusion Coefficient / Variance Rate ($D^{(2)}$ / Var)

What it is: Second-order Kramers-Moyal diffusion tensor measuring conditional variance rate of hidden-state increments.

Calculation:

$$D^{(2)}(z) = \lim_{\Delta t \rightarrow 0} \frac{1}{2\Delta t} \mathbb{E}[(\Delta z)(\Delta z)^\top \mid z_t = z]$$

In Words: "The second-order diffusion coefficient $D^{(2)}$ equals the limit as delta t approaches zero of one divided by two times delta t , multiplied by the expected outer product of hidden-state increments delta z with themselves."

Target / Healthy Values: Healthy range: 2.2×10^{-4} to 2.9×10^{-4} . Sudden jump $> 4.0 \times 10^{-4}$ indicates noise explosion.

Why it matters: Quantifies continuous stochastic perturbation acting on the representational trajectory.

Metric 2.3: Fourth-Order Kramers-Moyal Coefficient ($D^{(4)}$ / D_4)

What it is: Fourth-order stochastic moment auditing higher-order non-Gaussian jump noise.

Calculation:

$$D^{(4)}(z) = \lim_{\Delta t \rightarrow 0} \frac{1}{24\Delta t} \mathbb{E}[(\Delta z)^4 \mid z_t = z]$$

In Words: "The fourth-order Kramers-Moyal coefficient $D^{(4)}$ equals the limit as delta t approaches zero of one divided by twenty-four times delta t , multiplied by the expected fourth moment of increments delta z ."

Target / Healthy Values: Healthy range: 10^{-7} to 10^{-6} . Sustained values $> 5 \times 10^{-6}$ signal non-Langevin jump noise.

Why it matters: Tests whether the system obeys continuous Langevin dynamics or suffers from discontinuous state jumps.

Metric 2.4: Pawula Diagnostic Ratio ($\mathcal{R}_{\text{Pawula}}$)

What it is: Normalized Pawula ratio testing for Fokker-Planck truncation validity ($D^{(n)} = 0$ for $n \geq 3$).

Calculation:

$$\mathcal{R}_{\text{Pawula}} = \frac{D^{(4)}}{(V_{\text{Pawula}})^2}$$

In Words: "The Pawula ratio $\mathcal{R}_{\text{Pawula}}$ equals the fourth-order Kramers-Moyal coefficient $D^{(4)}$ divided by the square of the Pawula variance scale V_{Pawula} ."

Target / Healthy Values: Healthy range: 6.0 to 50.0. Values > 50.0 signal breakdown of continuous SDE assumptions.

Why it matters: By Pawula's Theorem, if $D^{(4)} \rightarrow 0$, all higher-order terms vanish, proving that trajectory evolution is governed strictly by drift and diffusion.

Metric 2.5: SDE Directional Signal-to-Noise Ratio (ρ_{dir})

What it is: Ratio of target-directed deterministic drift to effective diffusion noise intensity.

Calculation:

$$\rho_{\text{dir}} = \frac{\langle D^{(1)}(z), v_{\text{target}} \rangle}{\sqrt{D_{\text{eff}}^{(2)}(z)}}$$

In Words: "The directional signal-to-noise ratio ρ_{dir} equals the inner product of first-order drift vector $D^{(1)}$ and target direction vector v_{target} , divided by the square root of effective diffusion $D^{(2)}$."

Regime Thresholds:

- $\rho_{\text{dir}} > 100.0$: **Regime I (Persistent Individual Ballistic Flow - PIBF)**. Tokens travel along deterministic geodesics.
- $\rho_{\text{dir}} \ll 1.0$: **Regime II (Stochastic Collapse)**. Trajectories degenerate into random walks.

Why it matters: Identifies whether reasoning is driven by ballistic vector flow or collapsing into thermalized noise.

4 Latent Trajectory Kinematics & Differential Geometry

Metric 3.1: Latent Velocity ($v_t^{(l)}$)

What it is: Layer-to-layer vector displacement in the residual stream.

Calculation:

$$v_t^{(l)} = h_t^{(l)} - h_t^{(l-1)}, \quad \|v_t^{(l)}\|_2 = \sqrt{\sum_{i=1}^d (h_{t,i}^{(l)} - h_{t,i}^{(l-1)})^2}$$

In Words: "Latent velocity v at layer l equals the hidden state vector at layer l minus the hidden state vector at layer $l-1$."

Target / Healthy Values: $\|v\|_2$ range 2.8 to 4.2 units per layer.

Why it matters: Quantifies the magnitude of representational transformation performed by layer l .

Metric 3.2: Latent Acceleration ($a_t^{(l)}$)

What it is: Rate of change of latent velocity across consecutive network layers.

Calculation:

$$a_t^{(l)} = v_t^{(l)} - v_t^{(l-1)}$$

In Words: "Latent acceleration a at layer l equals velocity at layer l minus velocity at layer $l-1$."

Target / Healthy Values: $\|a\|_2$ range 0.4 to 1.2 units per layer².

Why it matters: Detects force application and momentum shifts in representational processing.

Metric 3.3: Trajectory Curvature (κ_t)

What it is: Angular curvature measuring how sharply the hidden-state trajectory bends through latent space.

Calculation:

$$\kappa_t = \frac{\|v_t \times a_t\|_2}{\|v_t\|_2^3 + \epsilon} \quad \text{or} \quad \kappa^{(l)} = \frac{\|v^{(l)}\|_2^2}{\|v^{(l)}\|_2^2 + \epsilon \|a^{(l)}\|_2}$$

In Words: "Curvature κ equals the norm of the cross product of velocity and acceleration, divided by the cubed norm of velocity plus epsilon."

Target / Healthy Values: Healthy range: 0.12 to 0.35. Spikes > 0.65 indicate logical reorientation or severe instability.

Why it matters: High curvature signals hesitation, context switching, or imminent trajectory breakdown.

Metric 3.4: Trajectory Jerk ($j^{(l)}$)

What it is: Third temporal derivative of position representing sudden, erratic changes in trajectory curvature.

Calculation:

$$j^{(l)} = \left| \kappa^{(l)} - \kappa^{(l-1)} \right|$$

In Words: "Jerk j at layer l equals the absolute difference between curvature at layer l and curvature at layer $l - 1$."

Target / Healthy Values: Bounded below 0.15. Spikes > 0.40 signal discontinuous shock updates.

Why it matters: Identifies abrupt, un-smoothed layer interventions that risk disrupting downstream representations.

Metric 3.5: Directional Persistence (I_t) & Rolling Persistence ($\bar{I}_t^{(k)}$)

What it is: Cosine alignment between consecutive velocity vectors, quantifying directional memory in latent motion.

Calculation:

$$I_t = \frac{v_t \cdot v_{t-1}}{\|v_t\|_2 \|v_{t-1}\|_2 + \epsilon}, \quad \bar{I}_t^{(k)} = \frac{1}{k} \sum_{i=t-k+1}^t I_i$$

In Words: "Persistence I at step t equals the dot product of velocity at step t and velocity at step $t - 1$, divided by the product of their Euclidean norms plus epsilon."

Regime Classification:

- $I_t \approx -1.0$: Anti-persistent (corrective, healthy exploratory reasoning).
- $I_t \approx 0.0$: Orthogonal directional reset.
- $I_t > 0.0$: Persistence flip crossing into attractor lock-in / repetitive collapse.

Why it matters: Primary early indicator of recursive collapse; persistent positive values ($I_t > 0.4$) indicate trapped, non-corrective looping.

Metric 3.6: Local Tension (σ_t)

What it is: Cross-run variance of rolling persistence across N independent random initializations or runs.

Calculation:

$$\sigma_t = \text{std} \left(\left\{ \bar{I}_t^{(r)} \right\}_{r=1}^N \right) = \sqrt{\frac{1}{N-1} \sum_{r=1}^N \left(\bar{I}_t^{(r)} - \langle \bar{I}_t \rangle \right)^2}$$

In Words: "Local tension σ at step t equals the sample standard deviation of rolling persistence across N independent runs."

Target / Healthy Values: Healthy range: 0.05 to 0.22. Spikes $\sigma_t > 0.35$ achieve AUROC > 0.9 in predicting collapse 2-3 tokens in advance.

Why it matters: Precursor stress metric; quantifies sensitivity to initial conditions prior to persistent flips.

Metric 3.7: Volatility Gradient (Φ_t) & Predictive Velocity ($\Psi_t^{(m)}$)

What it is: Discrete derivative of local tension ($\Phi_t = d\sigma_t/dt$) and its low-pass filtered momentum ($\Psi_t^{(m)}$) over window m .

Calculation:

$$\Phi_t = \sigma_t - \sigma_{t-1}, \quad \Psi_t^{(m)} = \frac{1}{m} \sum_{i=0}^{m-1} \Phi_{t-i}$$

In Words: "Volatility gradient Φ equals tension at step t minus tension at step $t - 1$. Predictive volatility velocity Ψ equals the moving average of Φ over window m ."

Target / Healthy Values: Φ_t bounded between -0.10 and $+0.10$.

Why it matters: Isolates structural stress acceleration from transient sampling noise.

Metric 3.8: Dubito Score / Paradox Index (Dubito)

What it is: Real-time paradox and uncertainty score measuring multi-head dissensus and trajectory curvature under contradictory inputs.

Calculation:

$$\text{Dubito}_t = \sigma_{\text{head},t}^2 \cdot \exp(\gamma \cdot \kappa_t) \cdot (1 - C_t)$$

In Words: "The Dubito score equals inter-head fiber curvature times the exponential of gamma times trajectory curvature, multiplied by one minus attention coherence C ."

Target / Healthy Values: Healthy range: 1.2 to 4.5. Paradox threshold: $\text{Dubito} > 7.0$.

Why it matters: Detects internal self-doubt and conflicting context coordinates before logit confidence collapses.

Metric 3.9: Effective Warning Lead Time (τ_{eff})

What it is: Time window (in tokens) between real-time telemetry anomaly detection and output token collapse.

Calculation:

$$\tau_{\text{eff}} = t_{\text{collapse}} - t_{\text{detection}}$$

In Words: "Effective warning lead time equals the timestep of token-level collapse minus the timestep of internal telemetry alert."

Target / Healthy Values: Target: $\tau_{\text{eff}} \geq +30.0$ tokens (MASSIF achieves mean $\tau_{\text{eff}} = +32.4$ tokens).

Why it matters: Demonstrates positive lead time over output probability metrics, enabling in-vivo intervention before bad tokens emit.

5 Friction, Depth Variance, & Silicon Hardware Rails

Metric 4.1: Early Layer Variance ($\mathcal{V}_{\text{early}}$) & Late Layer Variance ($\mathcal{V}_{\text{late}}$)

What it is: Mean normalized activation variance across early processing layers ($1 \rightarrow L/2$) versus late projection layers ($L/2 + 1 \rightarrow L$).

Calculation:

$$\mathcal{V}_{\text{early}} = \frac{1}{L/2} \sum_{\ell=1}^{L/2} \mathcal{V}^{(\ell)}, \quad \mathcal{V}_{\text{late}} = \frac{1}{L/2} \sum_{\ell=L/2+1}^L \mathcal{V}^{(\ell)}$$

In Words: "Early variance equals the average normalized activation variance over the first half of network layers; late variance equals the average over the second half."

Target / Healthy Values: $\mathcal{V}_{\text{early}} \in [0.93, 0.98]$; $\mathcal{V}_{\text{late}} \in [1.00, 1.06]$.

Why it matters: Tracks energy transfer from perceptual feature extraction to token generation.

Metric 4.2: Variance Delta / Friction Delta (Δ & Δ_{CFKR})

What it is: Depth-dependent variance differential measuring energy dissipation across layers.

Calculation:

$$\Delta = \mathcal{V}_{\text{early}} - \mathcal{V}_{\text{late}}$$

In Words: "Friction Delta equals mean early-layer variance minus mean late-layer variance."

Regime Classification:

- $\Delta > +1.0$: **DISSIPATED**. Early layers filter noise; late layers project freely.
- $-1.0 \leq \Delta \leq +1.0$: **HARMONIZED**. Healthy balanced depth variance ($\Delta \approx -0.06$ to -0.09).
- $\Delta < -1.0$: **DEEP DRIFT**. Late layers amplify noise; high runaway risk.
- $\Delta \rightarrow 0.00$: **Geometric Flatline**. Critical depth-specialization collapse triggering Cache-Fused Kinematic Rail (CFKR) intervention.

Why it matters: Triggers kernel-level KV-cache mutation in CFKR, achieving an $8.88\times$ speedup over software logit bias matrices.

6 Attention Consensus, Inter-Head Fiber Curvature, & Spectral Dynamics

Metric 5.1: Coherence Index (C) / Acclamation Rate

What it is: Fraction of token positions where cross-head attention variance falls below an adaptive threshold θ_t .

Calculation:

$$C = \frac{1}{T} \sum_{t=1}^T \mathbb{I}[\sigma_{\text{attn},t}^2 < \theta_t], \quad \theta_t = \text{Median}(\sigma_{\text{attn}}^2) + 1.4826 \cdot \text{MAD}(\sigma_{\text{attn}}^2)$$

In Words: "Coherence C equals the fraction of sequence steps where attention head variance is less than the MAD-scaled median threshold."

Target / Healthy Values: Healthy range: 0.72 to 0.88.

Why it matters: Measures structural consensus across attention heads.

Metric 5.2: Inter-Head Fiber Curvature (σ_{head}^2)

What it is: Variance of trajectory curvature across attention heads within a layer.

Calculation:

$$\sigma_{\text{head}}^2 = \text{Var}_{h \in \{1, \dots, H\}} \left(\kappa_t^{(h)} \right) = \frac{1}{H} \sum_{h=1}^H \left(\kappa_t^{(h)} - \bar{\kappa}_t \right)^2$$

In Words: "Fiber curvature σ_{head}^2 equals the sample variance of trajectory curvature across all H attention heads."

Target / Healthy Values: Healthy range: 0.60 to 0.77. Collapse below 0.50 indicates head homogenization.

Why it matters: Proxy for functional specialization across attention channels.

Metric 5.3: Decoupling Velocity ($\dot{\sigma}_{\text{head}}^2$)

What it is: First-order time derivative of inter-head fiber curvature.

Calculation:

$$\dot{\sigma}_{\text{head}}^2 \approx \frac{\sigma_{\text{head},t}^2 - \sigma_{\text{head},t-\Delta t}^2}{\Delta t}$$

In Words: "Decoupling velocity equals the rate of change of inter-head fiber curvature over time step Δt ."

Significance: $\dot{\sigma}_{\text{head}}^2 > 0$ with $\Delta\text{Loss} \approx 0$ confirms **Hidden Descent**—internal representational optimization occurring below scalar loss plateaus.

Metric 5.4: Block-4 Fourier Spectral Resonance ($S(\omega)$)

What it is: Power spectral density peak of hidden-state trajectories under periodic prompt forcing.

Calculation:

$$S(\omega) = \left| \frac{1}{T} \sum_{t=1}^T h_t e^{-i\omega t} \right|^2, \quad \omega_{\text{block4}} = \frac{2\pi}{4}$$

In Words: "Spectral density $S(\omega)$ equals the squared magnitude of the discrete Fourier transform of hidden state vectors at block frequency ω ."

Target / Healthy Values: Normalized resonance peak < 2.5 . Spikes > 6.0 indicate entrainment into "AAAABBBB" periodic loops.

Why it matters: Quantifies frequency-selective susceptibility to structural repetition in prompts.

7 Instability Forecasting & Closed-Loop Control

Metric 6.1: Integrated Instability Field ($I(l)$ & $I_{\text{combined}}(l)$)

What it is: Bayesian instability score combining consensus observation (P_{current}), historical persistence (P_{persist}), velocity, and acceleration.

Calculation:

$$I(l) = c(l)P_{\text{current}}(l) + (1 - c(l))P_{\text{persist}}(l), \quad I_{\text{combined}}(l) = I(l) + w_v \text{ReLU}(v_I) + w_a \text{ReLU}(a_I)$$

In Words: "Base instability I equals confidence times current prediction plus one minus confidence times historical persistence. Combined instability adds weighted velocity and acceleration terms."

Target / Healthy Values: Bounded below $I_{\text{target}} = 0.45$.

Why it matters: Provides a unified scalar field driving active governor interventions.

Metric 6.2: Forecast Confidence ($F(l)$)

What it is: Calibrated trust metric evaluating instability forecasts against coherence and curvature.

Calculation:

$$F(l) = \sigma(w_c C(l) + w_f F(l-1)) \cdot \exp(-\gamma \cdot \text{ReLU}(\kappa(l) - \kappa_0))$$

In Words: "Forecast confidence F equals the sigmoid of weighted coherence and historical confidence, exponentially damped by trajectory curvature exceeding threshold κ_0 ."

Target / Healthy Values: Operating range: 0.65 to 0.95.

Why it matters: Damps governor intervention strength when trajectory curvature is high to prevent over-correction during natural logical turns.

8 Pressure Tensor (Π) & Macroscopic Order Parameters

Metric 7.1: Global Optimization Pressure (Π)

What it is: Aggregate thermodynamic stress exerted by closed-loop governors on the residual stream, measured in Activation Magnitude Units (L_2 Norms).

Calculation:

$$\Pi = \frac{1}{L} \sum_{l=1}^L \sum_i \lambda_i(l) g_i(l) = \Pi_\alpha + \Pi_{\text{FFN}} + \Pi_{\text{MPC}} + \Pi_{\text{cap}}$$

In Words: "Global pressure Π equals the layer-averaged sum of governor work λ_i times constrained activation norm g_i across attention, FFN, MPC, and SoftCap channels."

Target / Healthy Values: Healthy aggregate range: 330.0 to 400.0 units.

Why it matters: Quantifies total internal stress required to maintain trajectory stability.

Metric 7.2: Constructive-Compensatory Ratio (R_t) [Primary Order Parameter]

, arc=2pt, top=6pt, bottom=6pt, left=8pt, right=8pt, boxrule=0.8pt ,options@for=metricbox] **What it is:** Macroscopic order parameter evaluating whether optimization is driven by structural attention learning or compensatory brute-force stabilization.

Calculation:

$$R_t = \frac{\Pi_\alpha}{\Pi_{\text{FFN}} + \Pi_{\text{MPC}}}$$

In Words: "The order parameter R_t equals attention governor pressure Π_α divided by the sum of feed-forward pressure Π_{FFN} and control pressure Π_{MPC} ."

Regime Classification:

- $R_t > 1.50$: **CONSTRUCTIVE**. Efficient attention-driven geometric learning.
- $1.15 \leq R_t \leq 1.50$: **MARGINAL**. Surfing the critical phase boundary.
- $R_t < 0.80$: **COMPENSATORY**. Glassy stasis / brute-force memorization.

Why it matters: Central state variable tracking representational maturity and phase transitions.

Metric 7.3: Optimization Response Function (χ_R)

What it is: Thermodynamic derivative of scalar loss with respect to macroscopic order parameter R_t .

Calculation:

$$\chi_R = \frac{\partial \mathcal{L}_{\text{loss}}}{\partial R_t} \approx \frac{\Delta \mathcal{L}_t}{\Delta R_t}$$

In Words: "The Optimization Response Function χ_R equals the partial derivative of loss with respect to the macroscopic order parameter R_t ."

Regime Sign Flips:

- $\chi_R < -0.05$: **Constructive Regime**. Increasing R_t minimizes loss.
- $\chi_R \approx 0.00$: **Critical Region** ($R_t \approx 1.2$). Order parameter and loss decouple.
- $\chi_R > +0.05$: **Compensatory Regime**. Increasing R_t increases loss (firefighting).

Why it matters: Acts as thermodynamic susceptibility, signaling when training enters diminishing returns.

9 Kuramoto Synchronization & Geometric Convergence (GCI)

Metric 8.1: Kuramoto Synchronization Order Parameter (R_t^{Kuramoto})

What it is: Ensemble order parameter quantifying directional phase alignment across N independent trajectories.

Calculation:

$$R_t^{\text{Kuramoto}} = \left\| \frac{1}{N} \sum_{r=1}^N u_t^{(r)} \right\|_2, \quad u_t^{(r)} = \frac{v_t^{(r)}}{\|v_t^{(r)}\|_2}$$

In Words: "The Kuramoto synchronization order parameter equals the norm of the ensemble average of normalized unit velocity vectors across N runs."

Target / Healthy Values: $R_t \approx 0$ indicates incoherent phase space; $R_t \approx 1$ indicates complete trajectory locking.

Why it matters: Measures global phase-locking across runs or attention heads.

Metric 8.2: Synchronization Velocity (V_t) & Acceleration (A_t)

What it is: First and second time derivatives of the Kuramoto synchronization parameter ($V_t = dR_t/dt, A_t = d^2R_t/dt^2$).

Dynamical Taxonomy Classes:

- **Accelerators** ($V_t > 0$): Synchronization builds before flip (GPT-2, Gemma, DeepSeek). Predictable warning.
- **Decelerators** ($V_t < 0$): Synchronization peaks then declines into flip (TinyLlama, RWKV).
- **Neutral** ($V_t \approx 0$): Flip occurs independent of synchronization (Qwen2).
- **Instant** (V_t undefined): Flips at step 1 with zero warning window (Bloom, Mamba).

Why it matters: Classifies model architectures into distinct dynamical vulnerability profiles.

Metric 8.3: Geometric Convergence Index (GCI) & Crystallization Horizon

What it is: Composite stopping metric auditing six disentangled information-geometric criteria to terminate pre-training at exact representational completion.

Calculation:

$$\text{GCI} = \frac{1}{6} \sum_{i=1}^6 \mathbb{I}[\text{Criterion}_i]$$

In Words: "GCI equals one-sixth times the sum of indicator functions across six criteria: (1) fiber curvature saturation, (2) decoupling velocity decay, (3) attention entropy variance stability, (4) governor work rate minimization, (5) Pawula stability ratio, and (6) macroscopic order parameter stability."

Phase Mapping:

- GCI = 0.00: **EARLY PLASTICITY**
- GCI = 0.50: **LATE DESCENT**
- GCI = 1.00 (sustained for 10,000 steps): **CRYSTALLIZATION HORIZON** (Stop Training).

Why it matters: Replaces scalar loss heuristics with a rigorous physics-grounded stopping standard, preventing over-training.

10 Advanced Anisotropic & Subspace Metrics

Metric 9.1: ANOVA Positional-Contextual Subspace Incoherence

What it is: Inner product metric measuring orthogonality between positional topology and contextual semantics under two-way ANOVA decomposition ($h = \mu + pos + ctx + resid$).

Calculation:

$$\text{Incoherence} = \max_{t,c} \left| \left\langle \frac{pos_t^{(l)}}{\|pos_t^{(l)}\|_2}, \frac{ctx_c^{(l)}}{\|ctx_c^{(l)}\|_2} \right\rangle \right|$$

In Words: "Subspace incoherence equals the maximum absolute inner product between normalized positional basis vector pos and normalized context basis vector ctx ."

Target / Healthy Values: Target: Incoherence < 0.15 .

Why it matters: Ensures clean orthogonal coordinate disentanglement between sequence position and semantic context.

Metric 9.2: Alpha Potential Well Regularizer ($U(\alpha)$)

What it is: Quadratic potential well acting as an effective Ricci curvature regulator against scaling symmetries under Manual Muon optimization.

Calculation:

$$U(\alpha) = \lambda_{\text{well}} \sum_{l=1}^L \left[\left(\alpha_{\text{attn}}^{(l)} - 1.0 \right)^2 + \left(\alpha_{\text{FFN}}^{(l)} - 1.0 \right)^2 \right]$$

In Words: "Potential energy $U(\alpha)$ equals scale factor λ_{well} times the sum across layers of squared deviations of attention and FFN layer scales from unity."

Target / Healthy Values: Bounded potential energy $U(\alpha) < 0.25$.

Why it matters: Suppresses scale drift in layer-norm parameters during long-run training.

11 Summary Matrix of Telemetry Observables

Field Name	Symbol	Mathematical Formula	Healthy Range / Target	Diagnostic Significance
Var	$D^{(2)}$	$\lim \frac{1}{2\Delta t} \mathbb{E}[\Delta z \Delta z^\top]$	$2.2 \times 10^{-4} - 2.9 \times 10^{-4}$	Second-order SDE variance rate
D4	$D^{(4)}$	$\lim \frac{1}{24\Delta t} \mathbb{E}[(\Delta z)^4]$	$10^{-7} - 10^{-6}$	Non-Gaussian jump noise check
Pawula Ratio	$\mathcal{R}_{\text{Pawula}}$	$D^{(4)}/(V_{\text{Pawula}})^2$	$6.0 - 50.0$	Fokker-Planck truncation validity
SDE SNR	ρ_{dir}	$\ D^{(1)}\ _2 / \sqrt{D^{(2)}}$	> 100.0	Ballistic reasoning vs random walk
Dubito Index	Dubito	$\sigma_{\text{head}}^2 e^{\gamma\kappa} (1 - C)$	$1.2 - 4.5$ (Threshold > 7.0)	In-vivo paradox / uncertainty alert
Lead Time	τ_{eff}	$t_{\text{collapse}} - t_{\text{detection}}$	$\geq +30.0$ tokens	Predictive warning window
Friction Delta	Δ	$\mathcal{V}_{\text{early}} - \mathcal{V}_{\text{late}}$	$-0.06 - -0.09$	Depth energy cascade / CFKR trigger
Fiber Curvature	σ_{head}^2	$\text{Var}_h(\kappa_t^{(h)})$	$0.60 - 0.77$	Attention head specialization
Decoupling Vel	$\dot{\sigma}_{\text{head}}^2$	$\Delta\sigma_{\text{head}}^2 / \Delta t$	$> +0.005$	Active Hidden Descent rate
Block-4 Peak	$S(\omega)$	$ \frac{1}{T} \sum h_t e^{-i\omega t} ^2$	< 2.5	Spectral prompt entrainment
Total Pressure	Π	$\Pi_\alpha + \Pi_{\text{FFN}} + \Pi_{\text{MPC}} + \Pi_{\text{cap}}$	$330.0 - 400.0$ units	Total activation stress
Order Param	R_t	$\Pi_\alpha / (\Pi_{\text{FFN}} + \Pi_{\text{MPC}})$	> 1.50 (Constructive)	Constructive vs Compensatory phase
Response Func	χ_R	$\partial\mathcal{L}/\partial R_t$	< -0.05	Thermodynamic susceptibility

Field Name	Symbol	Mathematical Formula	Healthy / Target Range	Diagnostic Significance
GCI Index	GCI	$\frac{1}{6} \sum \mathcal{K}[\text{Crit}_i]$	1.00 at Crystallization	Stopping criterion for pre-training